30/09/26

Tarea 5: Convolucion de Sistema LTI Discreto

Se tiene un sistema LTI con entrada $x[n]$ y respuesta al impulso unitario $h[n]$.

$$x[n] = \delta[n] + 2\delta[n-1] - \delta[n-3]$$

$$h[n] = 2\delta[n+1] + 2\delta[n-1]$$

Encontrar y graficar la respuesta $y[n]$ para las siguientes convoluciones:

(a) $y_1[n] = x[n] * h[n]$ (b) $y_2[n] = x[n+2] * h[n]$

(c) $y_3[n] = x[n] * h[n+2]$

a) $y_1[n] = (\delta[n] + 2\delta[n-1] - \delta[n-3]) * 2\delta[n+1] + 2\delta[n-1]$

$y_1[n] = x[n] * 2\delta[n+1] + 2\delta[n-1])$

$\delta[n] * 2\delta[n+1] = 2\delta[n+1]$

$\delta[n] * 2\delta[n+1] = 2\delta[n-1]$

$2\delta[n-1] * 2\delta[n+1] = 4\delta[n]$

$2\delta[n-1] * 2\delta[n-1] = 4\delta[n-2]$

$-\delta[n-3] * 2\delta[n+1] = -2\delta[n-2]$

$-\delta[n-3] * 2\delta[n-1] = -2\delta[n-4]$

$2x[n+1] = 2(\delta[n+1] + 2\delta[n] - \delta[n-2])$

$= 2\delta[n+1] + 4\delta[n] - 2\delta[n-2]$

$2x[n-1] = 2(\delta[n-1] + 2\delta[n-2] - \delta[n-4]$

$= 2\delta[n-1] + 4\delta[n-2] - 2\delta[n-4]$

$y_1[n] = 2\delta[n+1] + 4\delta[n] + 2\delta[n-1] + (-2+4)$

$\delta[n-2] - 2\delta[n-4]$

$y_1[n] = 2\delta[n+1] + 4\delta[n] + 2\delta[n-1] + 2\delta[n-2] -$

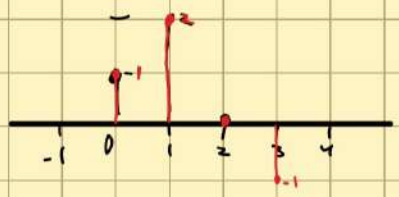
$2\delta[n-4]$

$y[-1] = 2$ $y[0] = 2$ $y[3] = 0$

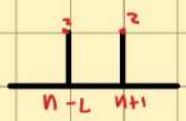
$y[0] = 4$ $y[2] = 2$ $y[4] = -2$

Graficar a)

Entrada $x[n] \rightarrow x[k]$



$h[n] \rightarrow h[k]$



Para $n < -1 \rightarrow y_1[n] = 0$

Para $n > 4 \rightarrow y_1[n] = 0$

$y_n[n]$ salida



b) $y_2[n] = x[n+2] * h[n]$

$y_2[n] = y_1[n+2]$ se recorre 2 unidades

$n \rightarrow n+2$

$2\delta[(n+2)+1] = 2\delta[n+3]$

$4\delta[n+2] = 4\delta[n+2]$

$2\delta[(n+2)-1] = 2\delta[n+1]$

$2\delta[(n+2)-2] = 2\delta[n]$

$-2\delta[(n+2)-4] = -2\delta[n-2]$

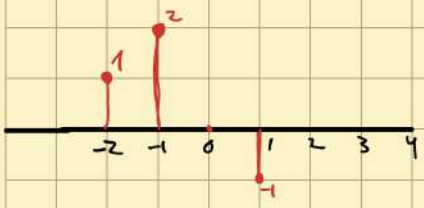
$y_2[n] = 2\delta[n+3] + 4\delta[n+2] + 2\delta[n+1] + 2\delta[n]$

$y[0] = 4$ $y[1] = 2$ $y[2] = 2$ $y[4] = -2$

Graficar b)

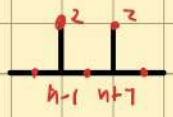
Entrada

$x[n+2] \rightarrow x[k+2]$



$h[n] \rightarrow h(n-k)$ Para $n < -3 \rightarrow Y_2[n] = 0$

$n > 2 \rightarrow Y_2[n] = 0$

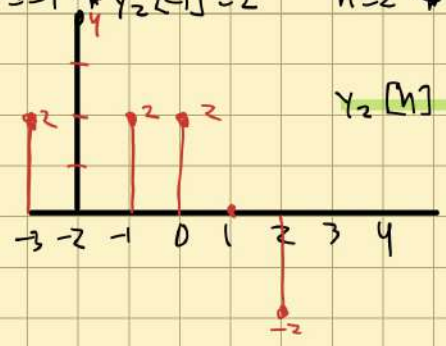


$n = -3 \rightarrow Y_2[-3] = 2$ $n = 0 \rightarrow Y_2[0] = 2$

$n = -2 \rightarrow Y_2[-2] = 4$ $n = 1 \rightarrow Y_2[1] = 0$

$n = -1 \rightarrow Y_2[-1] = 2$ $n = 2 \rightarrow Y_2[2] = -2$

$Y_2[n]$ salida



C) $Y_3[n] = x[n] + h[n+2]$

$Y_3[n] = Y_2[n] = Y_1[n+2]$

$Y_2[n] = 2\delta[n+3] + 4\delta[n+2] + 2\delta[n+1] + 2\delta[n+1]$

$+2\delta[n] - 2\delta[n-2]$

$n \rightarrow n+2$ de Y_1

$2\delta[(n+2)+1] = 2\delta[n+3]$

$4\delta[n+2] = 4\delta[n+2]$

$2\delta[(n+2)-1] = 2\delta[n+1]$

$2\delta[(n+2)-2] = 2\delta[n]$

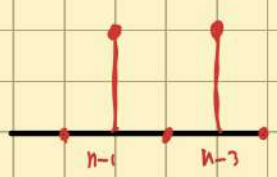
$-2\delta[(n+2)-4] = -2\delta[n-2]$

Graficar c)

Entrada $x[n] \rightarrow x[k]$



$h[n] \rightarrow h[n+2-k]$



Para $n < -3 \rightarrow Y_3[n] = 0$

Para $n \geq 2 \rightarrow Y_3[n] = 0$

$n = -3 \rightarrow Y_2[-3] = 2$ $n = 0 \rightarrow Y_2[0] = 2$

$n = -2 \rightarrow Y_2[-2] = 4$ $n = 1 \rightarrow Y_2[1] = 0$

$n = -1 \rightarrow Y_2[-1] = 2$ $n = 2 \rightarrow Y_2[2] = -2$

$Y_3[n]$ salida

